ChatGPT

Finite Continued-Fraction Control and Bounded-Digit Gaps: A Scholarly Novelty Audit

Executive summary

The mathematical mechanisms underlying Experiments 1 and 2 are already established, rigorously proved, and extensively peer-reviewed.

Experiment 1 interprets a finite regular continued-fraction prefix as an operator word that "steers" the location and "focuses" the width of the set of compatible infinite continuations. In classical terminology, the compatible set is a **continued-fraction cylinder**. Its endpoints and exact length are elementary functions of the convergents, and the denominator is an Euler continuant. Thus the facts that finite prefixes localize infinite tails, that cylinder lengths shrink with depth, and that digit ordering changes focusing are classical consequences of cylinder and continuant theory. The experiment's particular dispersion statistics are legitimate new computations, but the underlying phenomenon is not a new theorem. [1]

Experiment 2 restricts all permitted digits to ${1,\ldots,K}$ and finds persistent unreachable regions. The limiting reachable set is exactly the classical continued-fraction Cantor set

$$ E_K= \left\{ [0;a_1,a_2,\ldots]: 1\leq a_i\leq K \right\}. $$

For every finite $K\geq 2$, E_K is a compact, nowhere-dense conformal Cantor set with Hausdorff dimension $0<d_K<1$; consequently it has zero Lebesgue measure and permanent complementary gaps. Good, Cusick, Hensley, Mauldin–Urbański, Jenkinson–Pollicott, and others developed increasingly precise proofs, pressure formulas, transfer-operator methods, and rigorous algorithms for these sets. [2]

The exact experimental statement that, for $(K,m)=(5,12)$, depth 10 had already discovered the complete reachable grid mask is **not itself a published theorem located in this review**. It is a finite computational certificate about one grid. Its mathematical interpretation, however, is completely classical: a target interval is eventually reachable by bounded-digit prefixes exactly when its interior intersects E_K. The uploaded record reports exact enumeration, rational arithmetic, tripwires, and the observed saturation result, but it has not undergone external peer review. filecite turn0file0

The strongest defensible verdict is therefore:

Claim	Scholarly status
A finite prefix confines all infinite continuations to an exact interval	Completely solved; elementary cylinder theory
Prefix depth and digits determine cylinder position and width	Completely solved

Claim	Scholarly status
Reordering a fixed digit multiset can change focusing	Completely solved as a phenomenon; closely covered by continuant rearrangement theory
Bounded alphabets produce a fixed fractal reachable set	Completely solved
Missing regions can remain unreachable at every depth	Completely solved
The precise $K=5$, $m=12$, depth-10 mask is complete	Internally computationally certified, not independently peer-reviewed here
The particular dispersion and greedy-regret statistics	Apparently original descriptive computations, not foundational new mathematics
Cellwise quantities K_*, N_*, and O_*	Pointwise computability is straightforward; global scaling laws, complexity, and distributions remain plausible research territory

The project should **not proceed directly to a large Experiment 3 run**. The correct next step is a formal novelty audit and theorem-design phase focused on asymptotic laws, computational complexity, target-conditioned stopping times, and Pareto geometry rather than merely computing another finite heat map.

Abstract

This paper compares two computational experiments on regular continued fractions with the established literature. A finite word $w=(a_1,\ldots,a_n)$ defines a cylinder $\mathcal C_w$ containing every real number whose continued-fraction expansion begins with w. “Steering” means varying the cylinder’s position, while “focusing” means reducing its diameter. Restricting digits to $1\leq a_i\leq K$ produces the limit set E_K. We show that the principal experimental claims follow from exact cylinder formulas, continuant identities, conformal iterated-function-system theory, and thermodynamic formalism. In particular, $\lvert\mathcal C_w\rvert=1/[q_n(q_n+q_{n-1})]$, while eventual target reachability is equivalent to intersection with E_K. Classical and modern results prove that E_K is a Cantor set of dimension strictly below one for every finite K, and rigorous transfer-operator algorithms compute its dimension with certified error bounds. We then compare the proposed quantities $K_*(J)$, $N_*(J;K)$, and $O_*(J;K)$ with existing dimension, continuant, covering, renewal, and stopping-time theory. Their pointwise definitions do not by themselves constitute major open problems, but their target-conditioned distributions, asymptotic scaling, worst-case complexity, and Pareto structure do not appear to be resolved by the cited literature.

Introduction and background

Every irrational $x\in(0,1)$ has a regular continued-fraction expansion

$$ x=[0;a_1,a_2,\ldots], \qquad a_i\in\mathbb N. $$

The Gauss map

$$ T(x)=\frac{1}{x}-\left\lfloor\frac{1}{x}\right\rfloor $$

shifts the expansion, while the inverse branches

$$ \phi_a(x)=\frac{1}{a+x} $$

prepend the digit a. For a finite alphabet $A\subset\mathbb N$, the family ${\phi_a:a\in A}$ is a conformal iterated function system, and its limit set is precisely the set E_A of numbers whose continued-fraction digits all lie in A. For $A={1,\ldots,K}$, this is E_K. This symbolic/IFS representation is standard in modern treatments of restricted continued fractions. [3]

Historically, several related but distinct results must be separated. Jarník's 1928 work showed, among other early bounds, that the union of all bounded-type sets—the set of numbers whose partial quotients are bounded by some number depending on the point—has full Hausdorff dimension one. This does **not** say that any fixed E_K has dimension one. Jenkinson and Pollicott's historical summary records Jarník's lower bound $\dim_H(E_2)>1/4$, Good's refinement

$$ 0.5306<\dim_H(E_2)<0.5320, $$

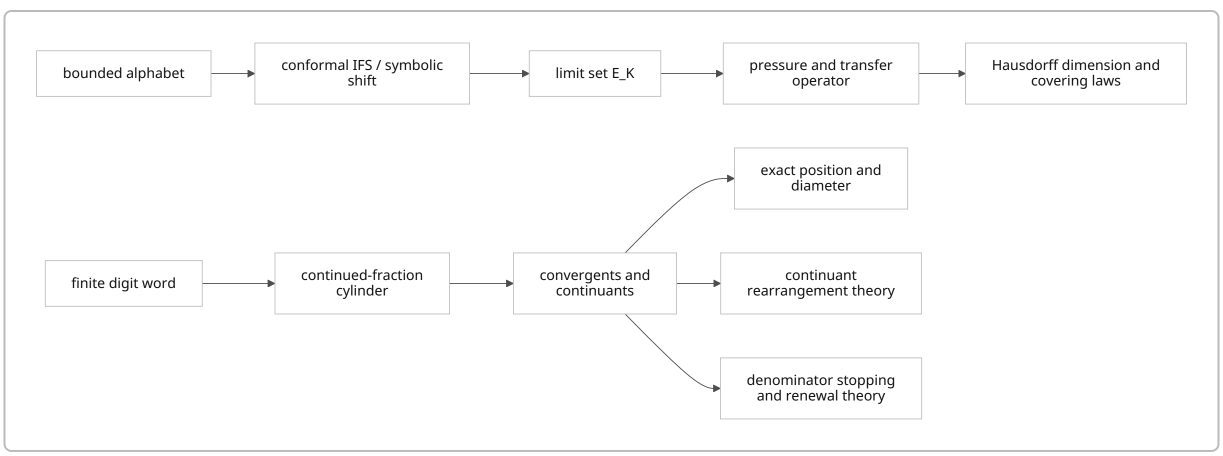
and Hensley's later rigorous enclosure

$$ 0.53128049<\dim_H(E_2)<0.53128051. $$

[4]

Good's 1941 paper systematically studied fractional dimensions of continued-fraction sets under restrictions on their partial quotients and explicitly treated them as zero-length exceptional sets. Hensley's papers from 1989 onward transformed this into a highly effective computational and functional-analytic theory. [5]

The relation among the principal theories is:



Experiment 1 lies principally in the top branch. Experiment 2 lies principally in the lower branch. Experiment 3 would combine the branches into a target-conditioned optimization problem.

Formal problem statements matching the experiments

Let $w=(a_1,\ldots,a_n)$ be a finite word. Define convergents by

$$ p_{-1}=1,\qquad p_0=0,\qquad q_{-1}=0,\qquad q_0=1, $$

and, for $j\geq1$,

$$ p_j=a_jp_{j-1}+p_{j-2}, \qquad q_j=a_jq_{j-1}+q_{j-2}. $$

The Möbius map represented by the word is

$$ F_w(t) = \phi_{a_1}\circ\cdots\circ\phi_{a_n}(t) = \frac{p_n+t p_{n-1}}{q_n+t q_{n-1}}. $$

The unrestricted-tail cylinder is

$$ \mathcal C_w=F_w([0,1]). $$

Its endpoints are

$$ \frac{p_n}{q_n} \quad\text{and}\quad \frac{p_n+p_{n-1}}{q_n+q_{n-1}}, $$

with their order determined by the parity of n. Since

$$ p_nq_{n-1}-p_{n-1}q_n=(-1)^{n-1}, $$

its exact length is

$$ \boxed{ \lvert\mathcal C_w\rvert = \frac{1}{q_n(q_n+q_{n-1})}. } $$

Thus the experiment's focusing score in bits can be written exactly as

$$ S(w) = -\log_2\lvert\mathcal C_w\rvert = \log_2 q_n+\log_2(q_n+q_{n-1}). $$

This identity resolves the basic "focusing" mechanism: focusing is controlled by adjacent continuants. The denominator q_n is the Euler continuant $K(a_1,\ldots,a_n)$, and the classical literature explicitly identifies continuants with continued-fraction denominators. [6]

Experiment 1 can therefore be stated precisely as the study, at fixed K and n, of the finite cloud

$$ \left{ \left(\operatorname{loc}(\mathcal C_w), -\log_2\lvert\mathcal C_w\rvert \right): w\in{1,\ldots,K}^n \right}, $$

where loc may be the midpoint, an endpoint, or another chosen location functional. The reported conditional dispersion statistics ask how much cylinder diameter varies among words whose locations fall in the same coarse cell, and conversely how broadly locations vary within a focusing band.

These statistics are not standard invariants in the literature, but their existence requires no new dynamical principle. Location depends on both numerator and denominator continuants, while length depends on q_n and q_{n-1}; there is no functional reason for coarse location to determine diameter.

For Experiment 2, define

$$ E_K = \bigcap_{n\geq1} \bigcup_{w\in{1,\ldots,K}^n} \mathcal C_w = \left{ [0;a_1,a_2,\ldots]: 1\leq a_i\leq K \right}. $$

For a target interval J with rational endpoints, define eventual robust reachability by

$$ \mathsf{Reach}_K(J) \iff \exists w\in{1,\ldots,K}^{<\infty} \text{ such that } \mathcal C_w\subseteq J. $$

The key equivalence is

$$ \boxed{ \mathsf{Reach}_K(J) \iff J^\circ\cap E_K\neq\varnothing. } $$

Proof sketch. If $\mathcal C_w\subseteq J$, append any infinite sequence of digits in ${1,\ldots,K}$. The resulting point belongs to $E_K\cap J$. Conversely, let $x\in E_K\cap J^\circ$. Its bounded-digit prefixes generate nested cylinders $\mathcal C_{w_n}$ containing x. Their denominators tend to infinity, so

$$ \lvert\mathcal C_{w_n}\rvert = \frac{1}{q_n(q_n+q_{n-1})} \longrightarrow0. $$

Because x lies in the interior of J, some cylinder is eventually contained in J.

This equivalence converts Experiment 2 from an empirical depth question into a fixed-set intersection problem. Depth does not create new asymptotic destinations; it refines the cylinder approximation to E_K.

Established theorems resolving Experiments 1 and 2

Cylinder localization and focusing

Finite-prefix localization is completely resolved by the preceding Möbius and determinant formulas. Every finite word maps the full tail interval to a rational cylinder, every child cylinder lies within its parent, and cylinder diameter decays at least exponentially because the denominators satisfy

$$ q_n\geq F_{n+1}, $$

where F_n is the Fibonacci sequence. Equality occurs for the all-ones word. Consequently, even the weakest alphabet $K=1$ focuses exponentially, though it only converges to the singleton

$$ E_1= \left\{ \frac{\sqrt5-1}{2} \right\}. $$

The stronger experimental observation—that words at a common depth and coarse location may differ greatly in width—is a finite combinatorial fact about the non-injectivity of the map from words to coarse location cells. It is computationally measurable but not a novel foundational theorem.

Order sensitivity and continuants

The order effect follows immediately because continuants are generally not symmetric under arbitrary digit permutation. For example,

$$ K(a_1,a_2,a_3)=a_1a_2a_3+a_1+a_3, $$

so interchanging an endpoint digit with the middle digit changes the value in general.

The stronger extremal problem—given fixed multiplicities of digits, which permutation maximizes or minimizes the denominator—has a substantial peer-reviewed literature. Ramharter's 1983 work gave explicit extremal arrangements for regular continuants, unique up to reversal under the stated framework; later papers refined and extended the combinatorics. [7]

A qualification is important. The experimental cylinder width is governed by

$$ q_n(q_n+q_{n-1}), $$

not by q_n alone. Ramharter's classical extremal-continuant theorem therefore settles denominator rearrangement more directly than it settles every exact cylinder-width permutation problem. The observed fact that ordering matters is entirely classical; a complete classification of width-extremizing permutations for every multiset is a narrower combinatorial question deserving a separate audit.

Permanent gaps

For $K\geq2$, the maps

$$ \phi_a(x)=\frac{1}{a+x}, \qquad 1\leq a\leq K, $$

have disjoint open images and satisfy the open-set condition. Their limit set E_K is a conformal Cantor set coded by the full shift on K symbols. General conformal IFS theory expresses its Hausdorff dimension as the unique zero of a pressure function. Mauldin and Urbański developed the pressure, conformal-measure, and Perron–Frobenius framework for finite and countable conformal IFSs. [8]

For the Gauss system, define the transfer operator

$$ (\mathcal L_{K,s}f)(x) = \sum_{a=1}^{K} \frac{1}{(a+x)^{2s}} f\left(\frac{1}{a+x}\right). $$

The dimension $d_K=\dim_H(E_K)$ is the unique value of s for which the spectral radius of $\mathcal L_{K,s}$ equals one, equivalently the unique zero of the associated pressure. Jenkinson and Pollicott state this characterization explicitly for restricted continued-fraction systems. [4]

Hensley proved that

$$ 0<d_K<1 \qquad\text{for every finite }K\geq2, $$

and studied the approach of d_K to one as $K\to\infty$. Since a subset of $\mathbb R$ containing an interval has Hausdorff dimension one, $d_K<1$ proves that E_K contains no interval. Because E_K is compact, it is nowhere dense; its complement therefore contains permanent open gaps at all depths. [9]

This proves the conceptual conclusion of Experiment 2:

$$ \boxed{ \text{For finite }K,\text{ deeper prefixes refine }E_K \text{ but cannot fill its complementary gaps.} } $$

Finite-grid saturation

Let $\mathcal J_m$ be the finite collection of dyadic target cells at resolution m. The asymptotic reachable mask is

$$ \mathcal M_{K,m} = \left\{ J\in\mathcal J_m: J^\circ\cap E_K\neq\varnothing \right\}. $$

Because dyadic endpoints are rational and E_K contains only irrationals, an unreachable closed dyadic cell is positively separated from E_K. Uniform cylinder contraction then implies that some finite level certifies its exclusion. Likewise, every reachable cell has a finite contained cylinder. Therefore, for each fixed (K,m), there exists a finite saturation depth

$$ n_{\mathrm{sat}}(K,m)<\infty. $$

This theorem is elementary once the compact IFS structure is recognized. What is not supplied by classical theory is the experiment's particular numerical assertion

$$ n_{\mathrm{sat}}(5,12)\leq10. $$

That claim depends on the exact computational certificate and remains a result of the uploaded implementation rather than an independently peer-reviewed literature result.

Literature and computational certification

The historical development is unusually complete: the geometry, dimension, computation, and ordering components have all been treated in major journals.

Source	Principal result relevant here	Method	What it settles
Jarník, 1928	Early dimension bounds; bounded-type numbers collectively have full dimension	Diophantine approximation and Hausdorff measure	Historical foundation, not fixed-K gap removal
Good, 1941	Fractional dimensions for continued fractions with digit restrictions; restricted sets have zero length	Covering and continued-fraction estimates	Establishes classicality of bounded/restricted-digit fractals [10]
Cusick, 1977–1978	Continuants with bounded digits; dimension characterized through continuant growth and Dirichlet-series methods	Continuant inequalities	Connects cylinder denominators and dimension [11]
Ramharter, 1983	Extremal arrangements of regular continuants under digit permutations	Combinatorial rearrangement inequalities	Settles major forms of the operator-order question [12]
Hensley, 1989	Rigorous bound $0.53128049<d_2<0.53128051$	Recursive transfer-operator approximation	Certified numerical dimension [13]
Hensley, 1990	Asymptotic counts and distributions of rationals with bounded digits	Analytic number theory and continuants	Counts finite bounded-digit words by denominator scale [14]
Hensley, 1992	Functional-analytic dimension theory and large-K asymptotics	Transfer operators	Global behavior of d_K [15]
Hensley, 1996	Polynomial-time dimension algorithm for finite alphabets, with $O(N^7)$ bit complexity under the paper's input model	Certified numerical functional analysis	Algorithmic computation of $\dim_H(E_A)$ [16]
Jenkinson–Pollicott, 2001	Super-exponentially convergent periodic-orbit method	Dynamical determinant	Fast high-precision dimension approximation [17]
Sinai–Ulcigrai, 2008	Limiting distribution of the first convergent denominator exceeding a threshold	Renewal theory and mixing of a special flow	Typical denominator overshoot, related but not identical to $N_$ *or* $O_$ [18]
Jenkinson–Pollicott, 2018	More than 100 rigorously certified decimal places for d_2	Hardy-space transfer operators and determinant tail bounds	Modern gold-standard numerical certification [19]

Source	Principal result relevant here	Method	What it settles
De Luca–Edson–Zamboni, 2023	Modern extension of extremal continuant arrangements	Combinatorics and interval-exchange codings	Clarifies what is and is not solved in rearrangement theory [20]

For E_2, the rigorously established value begins

$$ \dim_H(E_2) = 0.531280506277205141624468647368\ldots, $$

with more than 100 decimal digits certified by explicit transfer-operator and determinant error bounds. [21]

The dimension trend is therefore not an empirical conjecture:

Alphabet	Hausdorff dimension
${1}$	0
${1,2}$	$0.531280506277205\ldots$
${1,\ldots,K}$, finite K	Strictly between 0 and 1
$K\to\infty$	$d_K\to1$

Hensley's transfer-operator work supplies refined asymptotics for the final row, while Urbański and Zdunik proved continuity behavior for the corresponding Hausdorff measures as the finite alphabets exhaust the full Gauss system. [22]

For covering numbers, standard finite conformal-IFS theory implies that the logarithmic covering exponent agrees with d_K:

$$ \lim_{r\downarrow0} \frac{\log N_r(E_K)}{-\log r} = d_K, $$

under the regularity and separation present here. Thus the leading exponent governing the number of resolution-r cells required to cover the reachable set is settled. Exact finite-grid counts, second-order oscillations, saturation depths, and alignment effects with a prescribed dyadic grid are much more specific quantities. A 2026 survey by Rutar treats multiscale restricted continued-fraction sets and covering behavior, but it is a recent preprint rather than part of the older peer-reviewed foundation. [23]

Experiment 3 novelty audit and genuine open problems

The proposed quantities are

$$ K_*(J) = \min\left\{ K:J^\circ\cap E_K\neq\varnothing \right\}, $$

$$ N_*(J;K) = \min\left\{ |w|: w\in\{1,\ldots,K\}^{<\infty}, \mathcal C_w\subseteq J \right\}, $$

and

$$ O_*(J;K) = \min_{\mathcal C_w\subseteq J} \log_2\frac{|J|}{|\mathcal C_w|}. $$

The sign convention above ensures $O_*\geq0$.

These quantities should not all be labeled “open problems” in the same sense. Their **pointwise existence and exact computability** are substantially easier than their **global laws**.

Quantity	Relation to established literature	What is already effectively solved	Plausibly open or underdeveloped
$K_*(J)$	Nested continued-fraction Cantor sets $E_1\subset E_2\subset\cdots$; bounded-type filtrations	For a rational interval, exact branch-and-bound can certify intersection or separation for each K; $K_*(J)$ is finite for every nonempty open J	Distribution over dyadic cells, sharp boundary asymptotics near 0, regularity of the landscape $J\mapsto K_*(J)$, and complexity as resolution grows
$N_*(J;K)$	Cylinder stopping times, coding depth, denominator growth, covering theory	Breadth-first exact search terminates whenever $J^\circ\cap E_K\neq\varnothing$; monotonicity in K is immediate	Typical and worst-case scaling with m, concentration laws, singular behavior near gaps, and complexity classes for exact computation
$O_*(J;K)$	Cylinder diameter thresholds and renewal overshoot	Exact branch-and-bound is possible because descendants are narrower and only finitely many cylinders exceed a fixed width	Limiting distribution across targets, existence of uniformly bounded overshoot away from gaps, and characterization of forced large overshoot
Pareto set $\mathcal P(J)$	Multiobjective coding and resource optimization	Finite nondominated sets can be computed below any relevant width/depth cutoff	Asymptotic shape, universality, phase transitions, and bounds on Pareto-front cardinality
Greedy regret	Search algorithms on symbolic trees	Exact optimum gives a benchmark	Approximation guarantees, impossibility results, adversarial target families, and optimal lookahead policies

For $K_*(J)$, a useful reformulation is

$$ K_*(J) = \min_{x\in J^\circ\setminus\mathbb Q} \sup_{n\geq1}a_n(x). $$

This is a local minimization of the bounded-type constant over an interval. The nested sets E_K and related bounded-type filtrations are classical, and modern work studies parameterized bounded-type sets and their bifurcations. However, the particular dyadic-cell threshold landscape does not appear as a standard named invariant in the sources reviewed here. [24]

For N_*, the cylinder formula gives the approximate denominator threshold

$$ q_n(q_n+q_{n-1}) \gtrsim |J|^{-1}. $$

For a target of width 2^{-m}, one expects denominators of order $2^{m/2}$ before containment is possible, but denominator size alone does not ensure spatial placement. Classical metric theory, Lévy-type growth, and renewal theory describe typical denominator growth along an already chosen continued-fraction orbit. Sinai and Ulcigrai prove a limiting law for the first q_n exceeding R, including the ratio q_n/R. Their theorem does **not** optimize over all bounded-digit words whose cylinders fit a prescribed spatial target. [25]

Similarly, O_* *is related to renewal overshoot but is not the same object. Renewal theory asks how far a typical orbit's denominator passes a threshold.* O_* asks for the widest admissible cylinder, among every bounded-digit word at every depth, that lies at a specified location. It combines a scale constraint with a spatial and alphabet constraint. This target-conditioned optimization appears to be the most promising genuinely distinct component.

Several research questions therefore remain credible:

Candidate question	Assessment
Does $N_*(J;K)/m$ converge in distribution when J ranges over reachable dyadic cells of width 2^{-m}?	Plausibly open; not resolved by ordinary renewal theory
Are there constants c_K,C_K such that most reachable cells satisfy $c_Km\leq N_*\leq C_Km$?	Likely approachable through pressure and large deviations
Is $O_*(J;K)$ uniformly bounded on cells a fixed relative distance from complementary gaps?	Nontrivial and potentially publishable
Can O_* be arbitrarily large for cells approaching particular gap endpoints?	Strong candidate for exact theorem
What is the worst-case computational complexity of exact K_*, N_*, and O_* as functions of endpoint bit length and m?	Apparently underdeveloped
Which permutations extremize the full cylinder-width functional $q_n(q_n+q_{n-1})$ for a prescribed digit multiset?	Narrow combinatorial question adjacent to, but not identical with, classical continuant extrema

Candidate question	Assessment
Can greedy synthesis achieve bounded additive or multiplicative regret over all reachable cells?	Algorithmic open direction
Do normalized Pareto fronts converge to a deterministic shape or multifractal family?	Speculative but genuinely new framing

These questions should be treated as **candidate open problems pending a targeted bibliography search**, not as definitively unsolved theorems merely because the exact notation is new.

Recommended research agenda and primary sources

The immediate research agenda should be theorem-first rather than experiment-first.

First, formalize the exact target model: open versus closed J, unrestricted-tail versus bounded-tail cylinders, handling of rational endpoints, and the sign of the overshoot definition. Prove finite termination and correctness of exact algorithms for K_*, N_*, and O_*. These are baseline propositions, not experimental discoveries.

Second, complete a focused novelty audit in four adjacent literatures: local hitting and shrinking-target problems for the Gauss map; optimal coding and shortest-cylinder problems in symbolic dynamics; renewal theorems for bounded-alphabet subshifts and Gibbs measures; and algorithmic complexity of interval–Cantor-set intersection. The ordinary Sinai–Ulcigrai theorem should be treated as a comparison theorem, not as a solution to the target-conditioned problem. [18]

Third, select one theorem-scale objective before computing full heat maps. The strongest options are a scaling theorem for N_*, *a forced-overshoot theorem for* O_* near gap boundaries, a worst-case complexity result, or a full rearrangement theorem for the cylinder-width functional.

Fourth, reproduce a small set of published benchmarks—especially d_2 and pressure roots—with rigorous interval arithmetic. Matching Jenkinson–Pollicott or Hensley would validate the implementation against an external mathematical gold standard rather than only internal tripwires. [26]

Only after these steps should a computational Experiment 3 be run. Its purpose should be to test a theorem-driven conjecture, not merely to visualize K_*, N_*, and O_*.

The most important primary sources are:

Primary source	Relevance
I. J. Good, “The Fractional Dimensional Theory of Continued Fractions,” *Proceedings of the Cambridge Philosophical Society* 37 (1941), 199–228	Foundational restricted-digit dimension theory. [10]

Primary source	Relevance
T. W. Cusick, “Continuants with Bounded Digits,” *Mathematika* 24 (1977), 166–172	Continuants, denominators, and bounded-digit geometry. [27]
G. Ramharter, “Extremal Values of Continuants,” *Proceedings of the AMS* 89 (1983), 189–201	Extremal digit orderings for regular continuants. [28]
D. Hensley, “The Hausdorff Dimensions of Some Continued Fraction Cantor Sets,” *Journal of Number Theory* 33 (1989), 182–198	Rigorous computation of d_2. [13]
D. Hensley, “Continued Fraction Cantor Sets, Hausdorff Dimension, and Functional Analysis,” *Journal of Number Theory* 40 (1992), 336–358	Transfer operators and asymptotic dimension theory. [15]
D. Hensley, “A Polynomial Time Algorithm for the Hausdorff Dimension of Continued Fraction Cantor Sets,” *Journal of Number Theory* 58 (1996), 9–45	Certified algorithmic dimension computation. [16]
R. D. Mauldin and M. Urbański, “Dimensions and Measures in Infinite Iterated Function Systems,” *Proceedings of the London Mathematical Society* 73 (1996), 105–154	General conformal IFS pressure and measure theory. [8]
O. Jenkinson and M. Pollicott, “Computing the Dimension of Dynamically Defined Sets: E_2 and Bounded Continued Fractions,” *Ergodic Theory and Dynamical Systems* 21 (2001), 1429–1445	Periodic-orbit and determinant method. [17]
Y. Sinai and C. Ulcigrai, “Renewal-Type Limit Theorem for the Gauss Map and Continued Fractions,” *Ergodic Theory and Dynamical Systems* 28 (2008), 643–655	Denominator stopping-time and overshoot law. [18]
O. Jenkinson and M. Pollicott, “Rigorous Effective Bounds … A Hundred Decimal Digits for the Dimension of E_2,” *Advances in Mathematics* 325 (2018), 87–115	State-of-the-art rigorous computation. [29]
A. De Luca, M. Edson, and L. Q. Zamboni, “Extremal Values of Semi-Regular Continuants and Codings of Interval Exchange Transformations,” *Mathematika* 69 (2023), 432–457	Modern account of what continuant rearrangement theory resolves. [20]

The final conclusion is precise:

$$ \boxed{ \begin{aligned} \text{Experiment 1’s mechanism} &: \text{classical and solved},\\ \text{Experiment 2’s permanent gaps} &: \text{classical and solved},\\ \text{the exact finite numerical tables} &: \text{computational results, not yet peer-reviewed},\\ \text{Experiment 3’s pointwise definitions} &: \text{computable but not inherently novel},\\ \text{their asymptotic resource geometry} &: \text{the principal plausible research opportunity}. \end{aligned} } $$

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https://arxiv.org/abs/1611.09276

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